

Micro Energy Harvesting

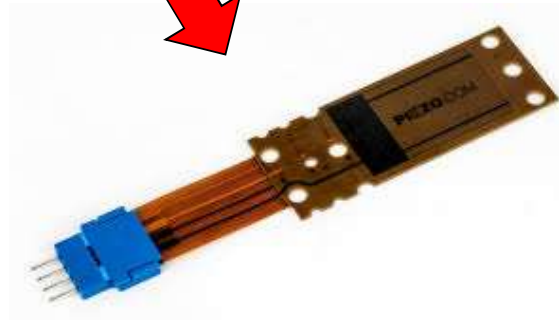
Vibrational Energy Harvesting

Vibrational Energy Harvesting - Concept



- You need some local sensor with low power consumption, but you don't want a cable.
- Solution: Harvest some energy from the vibration

- Marine diesel engine (up to 2300 t)
- You feel the vibration when you are on a ferry.
- There is a huge amount energy wasted in the vibration.



- Piezoelectric energy harvester (about 1 g)
- There is only a small amount of energy stored in the low mass of the harvester

Vibrational Energy Harvesting - Concept

Strategy for vibrational energy harvesting

- Energy stored in a vibrating mass: $\sim m \cdot x_0^2$
- $\frac{m_{\text{marine diesel engine}}}{m_{\text{harvester}}} \approx 10^9 \rightarrow$ no influence of harvester on the energy source (infinite energy reservoir)
- $m_{\text{harvester}}$ is typically limited (size or cost constraints)
- Try maximize the vibration amplitude $x_0 \rightarrow$ exploit resonance

*Nov. 1940: Tacoma Narrows Bridge,
collapsed after 4 month of serving due to resonance*



Vibrational Energy Harvesting - Concept

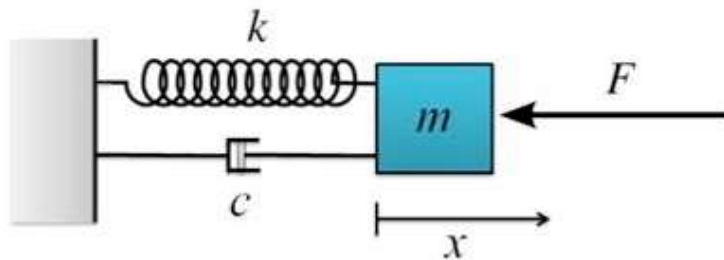
Content:

- Mathematical treatment of vibrations
- How to maximize the vibrational amplitude
- Condition for maximum power output
- How to find the right harvester design
- Piezoelectric harvesters
- Electrostatic harvesters
- Electromagnetic harvesters

The harmonic oscillator

The harmonic oscillator is a key model for many effects in nature and technology. Therefore it is always a good idea to refresh your knowledge of this model.

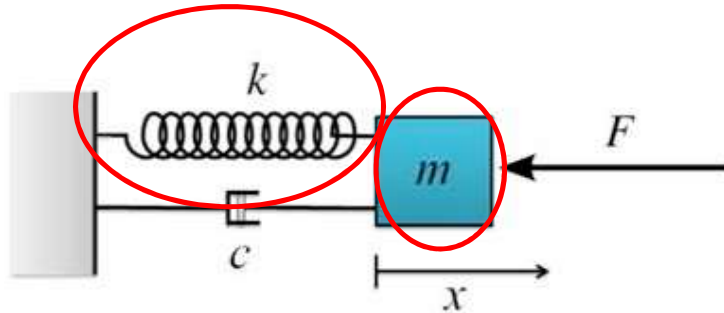
Lets start with a mechanical representation of this model:



We treat the following problems:

- undamped, unforced harmonic oscillator (only considering mass m and spring constant K)
- damped, unforced harmonic oscillator (only considering mass m and spring constant K and the viscous damping coefficient c)
- damped, forced harmonic oscillator (considering all elements of the graph)

The undampend unforced harmonic oscillator (1) - Solution



Solution:

$$x(t) = x_0 \cdot \cos \omega_0 t$$

$$\omega_0 = \sqrt{\frac{K}{m}}$$

Balance of forces:

$$-m \cdot \ddot{x}(t) - K \cdot x(t) = 0$$

Initial condition:

$$x(0) = x_0; \quad \dot{x}(0) = 0$$

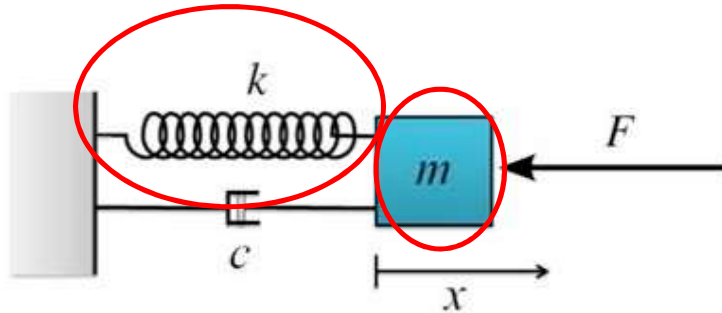
$$f_0 = 2\pi\omega_0, \quad T = \frac{1}{f_0} = \frac{2\pi}{\omega_0}$$

$$A_{pp} = 2 \cdot x_0$$

What did you learn?

If you displace the mass by x_0 from its equilibrium position it will oscillate with a constant **amplitude** of x_0 at its **natural frequency** f_0 .

The undampend unforced harmonic oscillator (2) - Energy



Total energy:

$$\begin{aligned} W_{tot} &= W_{kin} + W_{pot} \\ &= \frac{1}{2} K \cdot x_0^2 \cdot \cos^2 \omega_0 t + \frac{1}{2} K \cdot x_0^2 \cdot \sin^2 \omega_0 t \\ &= \frac{1}{2} K \cdot x_0^2 = \frac{1}{2} m \cdot \omega_0^2 \cdot x_0^2 = \text{const.} \end{aligned}$$

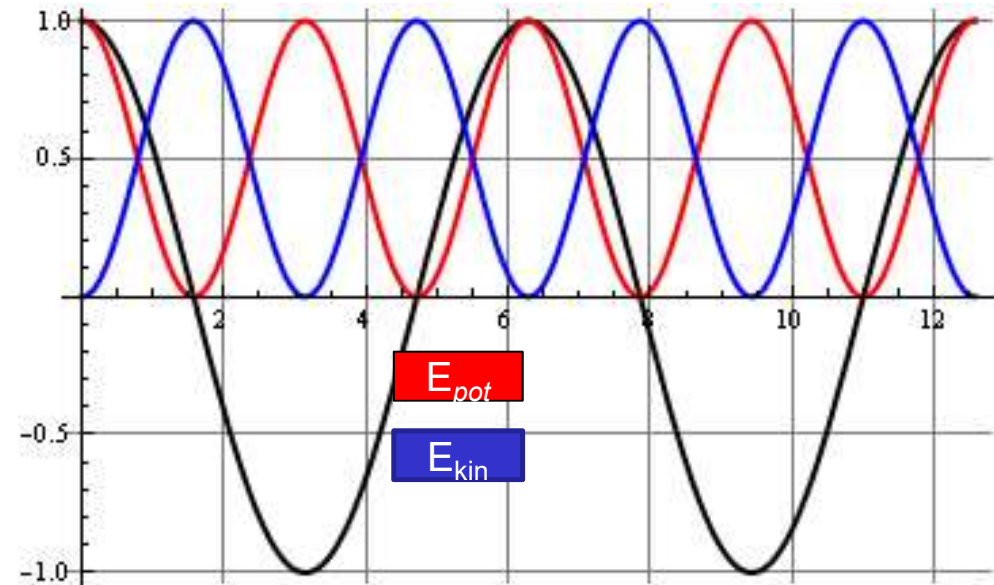
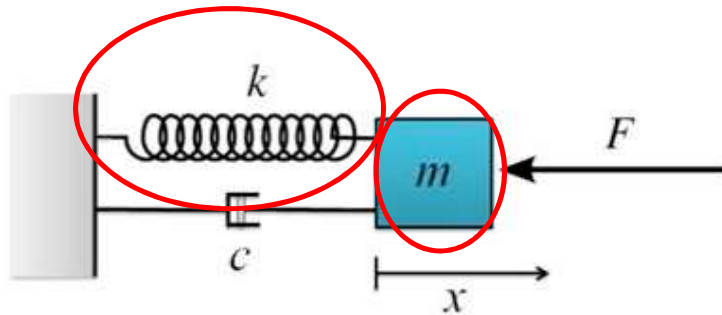
Kinetic energy:

$$\begin{aligned} W_{kin} &= \frac{1}{2} m \cdot \dot{x}^2 = \frac{1}{2} m \cdot \overbrace{\omega_0^2}^{\omega_0^2 = \frac{K}{m}} \cdot x_0^2 \cdot \sin^2 \omega_0 t \\ &= \frac{1}{2} K \cdot x_0^2 \cdot \sin^2 \omega_0 t \end{aligned}$$

Potential energy:

$$W_{pot} = \frac{1}{2} K \cdot x^2 = \frac{1}{2} K \cdot x_0^2 \cdot \cos^2 \omega_0 t$$

The undamped unforced harmonic oscillator (3) - Energy

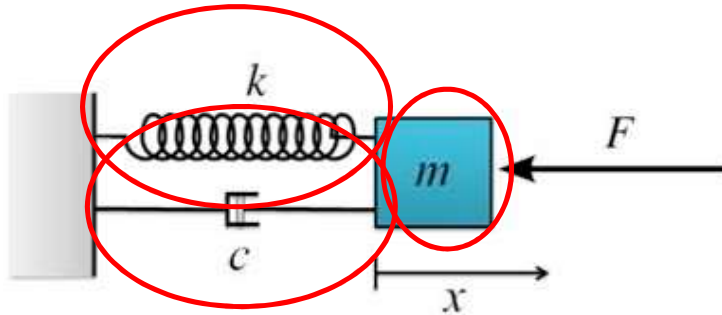


What did you learn?

The total energy of the system does not change.
The total energy oscillates between kinetic and potential energy but no exchange of energy with the surrounding takes place.

$$W_{tot} = \frac{1}{2} m \cdot \omega_0^2 \cdot x_0^2$$

The damped unforced harmonic oscillator (1) - Solution



Balance of forces: *Damping coefficient [kg/s]*

$$-m \cdot \ddot{x}(t) - c \cdot \dot{x}(t) - K \cdot x(t) = 0$$

Initial condition:

$$x(0) = x_0; \quad \dot{x}(0) = 0$$

Solution:

$$x(t) = x_0 \cdot e^{-\zeta \cdot \omega_0 t} \cos \omega_R t \cdot$$

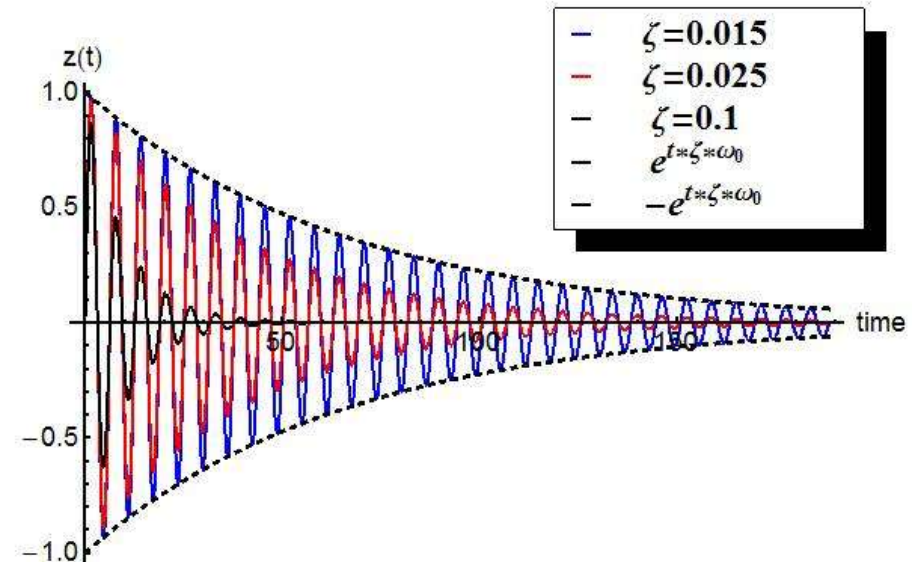
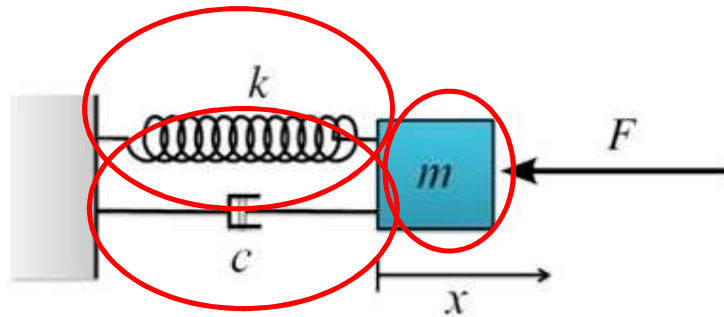
$$\zeta = \frac{c}{2m\omega_0} \quad \omega_R = \omega_0 \sqrt{1 - \zeta^2}$$

*Damping ratio
(dimensionless)*

resonance frequency

natural frequency

The damped unforced harmonic oscillator (2) - Solution



What did you learn?

If you displace the mass by x_0 from its equilibrium position it will oscillate with an **exponentially decaying amplitude** at its **resonance frequency** f_R which is near the **natural frequency** f_0

The damped unforced harmonic oscillator (3) - Energy

$$\begin{aligned}W_{kin} &= \frac{1}{2} m \cdot \dot{x}^2 = \dots \\&= \frac{1}{2} K \cdot x_0^2 \cdot e^{-2\zeta \cdot \omega_0 t} \cdot \left(\frac{\omega_0^2}{\omega_R^2} \cdot \sin^2 \omega_R t \right) \\&\approx \frac{1}{2} K \cdot x_0^2 \cdot e^{-2\zeta \omega_0 t} \cdot \sin^2 \omega_R t\end{aligned}$$

$$\begin{aligned}W_{pot} &= \frac{1}{2} K \cdot x^2 = \dots \\&= \frac{1}{2} K \cdot x_0^2 \cdot e^{-2\zeta \cdot \omega_0 t} \cdot \left(\cos \omega_R t + \zeta \frac{\omega_0}{\omega_R} \sin \omega_R t \right)^2 \\&\approx \frac{1}{2} K \cdot x_0^2 \cdot e^{-2\zeta \omega_0 t} \cdot \cos^2 \omega_R t\end{aligned}$$

The damped unforced harmonic oscillator (4) - Energy

$$\begin{aligned} W_{tot} &= W_{kin} + W_{pot} = \dots \\ &= \frac{1}{2} K \cdot x_0^2 \cdot e^{-2\zeta\omega_0 t} \\ &\cdot \left(\frac{\omega_0^2}{\omega_R^2} \cdot \sin^2 \omega_R t + \left(\cos \omega_R t + \zeta \cdot \frac{\omega_0}{\omega_R} \sin \omega_R t \right)^2 \right) \\ &\approx \frac{1}{2} K \cdot x_0^2 \cdot e^{-2\zeta\omega_0 t} \quad \text{with } \zeta \ll 1 \text{ and } \omega_R \approx \omega_0 \end{aligned}$$

What did you learn?

The total energy of the system decays with an exponential rate of $2\zeta\omega_0$.

The energy is dissipated (lost) in the damper.

The damped unforced harmonic oscillator (5) - Power

$$\begin{aligned} P_{damp} &= F_{damp} \cdot \dot{x}(t) = -c \cdot \dot{x}(t)^2 \\ &= -c \cdot x_0^2 \cdot e^{-2\zeta\omega_0 t} (-\gamma \cdot \cos\omega_R t + \omega_R e^{-\gamma \cdot t} \cdot \sin\omega_R t)^2 \\ &\approx -c \cdot x_0^2 \cdot e^{-2\zeta\omega_0 t} \cdot \omega_0^2 \cdot \sin^2 \omega_R t \end{aligned}$$

What did you learn?

The instantaneous power dissipated in the damper decreases with a rate of $2\zeta\omega_0$.

The energy is dissipated (lost) in the damper.

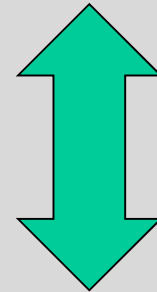
The damped unforced harmonic oscillator (6) - Energy

*Energy
stored in the
spring*

$$W_{in} = 1/2 K x_0^2$$

1. *Energy is stored in the spring*

*Energy stored
in the spring*



*Energy stored
the moving mass*

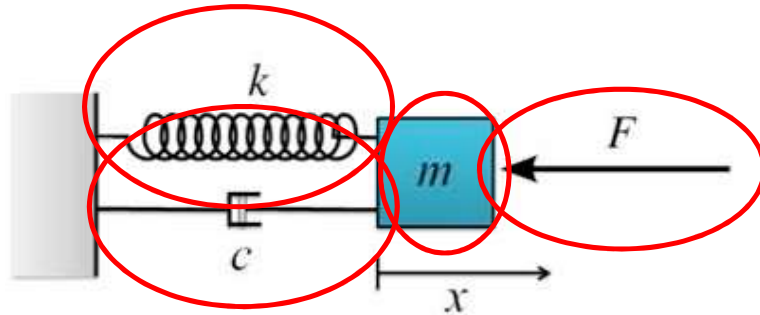


*Dissipated
energy*

$$W_{diss} = 1/2 K x_0^2$$

2. *If released the energy oscillates between the potential and kinetic reservoir. A small portion of the energy is dissipated in the damper, so the oscillation slowly dies out. At the end the total energy is dissipated.*

The damped forced harmonic oscillator (1) - Solution



Solution: ***Ugly looking equations...***

Balance of forces:

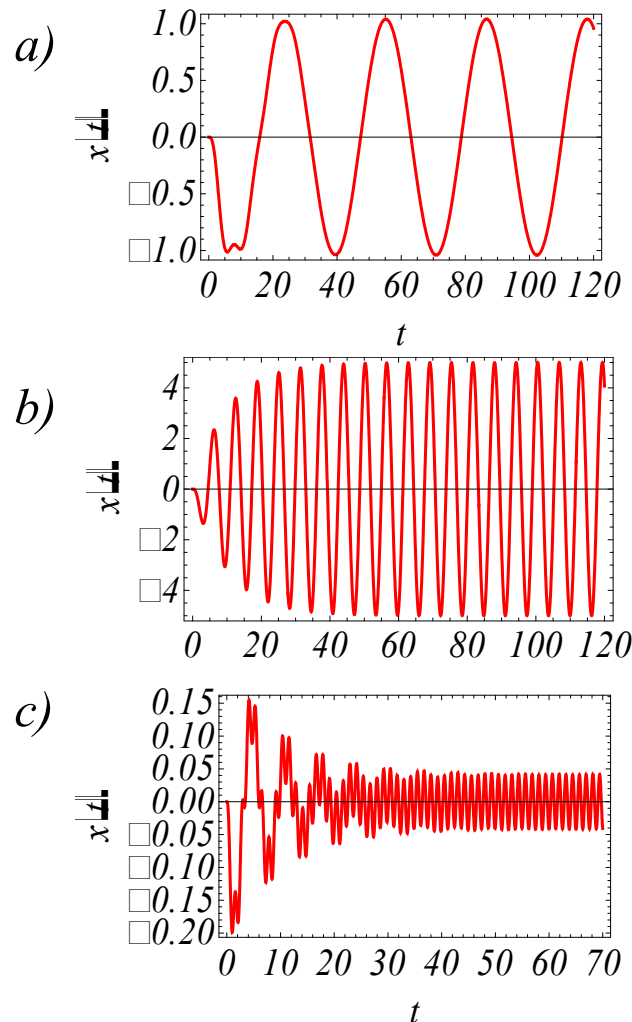
$$-m \cdot \ddot{x}(t) - c \cdot \dot{x}(t) - K \cdot x(t) = F_0 \cdot \cos \omega t$$

$$-\ddot{x}(t) - 2\zeta \cdot \omega_0 \cdot \dot{x}(t) - \omega_0^2 \cdot x(t) = \frac{F_0}{m} \cdot \cos \omega t$$

Initial condition:

$$x(0) = x_0; \quad \dot{x}(0) = 0$$

The damped forced harmonic oscillator (2) - Solution



$\omega_0=1$, $F_0/m=1$, $\zeta = 0.1$ for all graphs

a) $\omega = 0.2$ i.e. $\omega \ll \omega_0$

After a few cycles the displacement oscillates with the frequency ω , amplitude $F_0/m\omega_0^2$ in phase with the forcing

b) $\omega = 1$ i.e. $\omega = \omega_0$

After a few cycles the displacement oscillates with the frequency ω , amplitude $Q \cdot F_0/m\omega_0^2$ with a phase shift of 90° to the forcing.

c) $\omega = 5$ i.e. $\omega \gg \omega_0$

After a few cycles the displacement oscillates with the frequency ω , an amplitude much smaller than $F_0/m\omega_0^2$ with a phase shift of 180° to the forcing.

The damped forced harmonic oscillator (3) - Solution

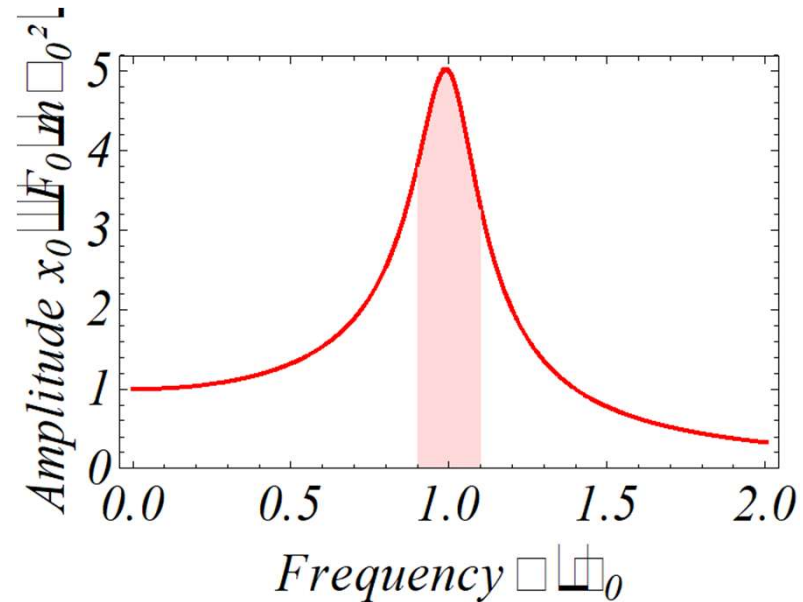
Actually we are only interested in the long time behaviour, typically also the phase of the displacement is of minor importance. Therefore we just need a formula for the amplitude x_0 in dependence on the forcing frequency ω .

$$x_0 = \frac{F_0}{m} \cdot \frac{1}{\sqrt{(\omega^2 - \omega_0^2)^2 + 4\zeta^2 \omega^2 \omega_0^2}}$$

$$\zeta \ll 1$$

$$\Rightarrow x_{0,Max} \approx x_0(\omega_0) = \frac{F_0}{m} \cdot \frac{1}{2\zeta \omega_0^2} = Q \cdot \frac{F_0}{m\omega_0^2}$$

$$x_0\left(\omega_0 \pm \frac{1}{2Q} \omega_0\right) \approx \frac{1}{\sqrt{2}} Q \cdot \frac{F_0}{m\omega_0^2} = \frac{1}{\sqrt{2}} \cdot x_{0,Max}$$



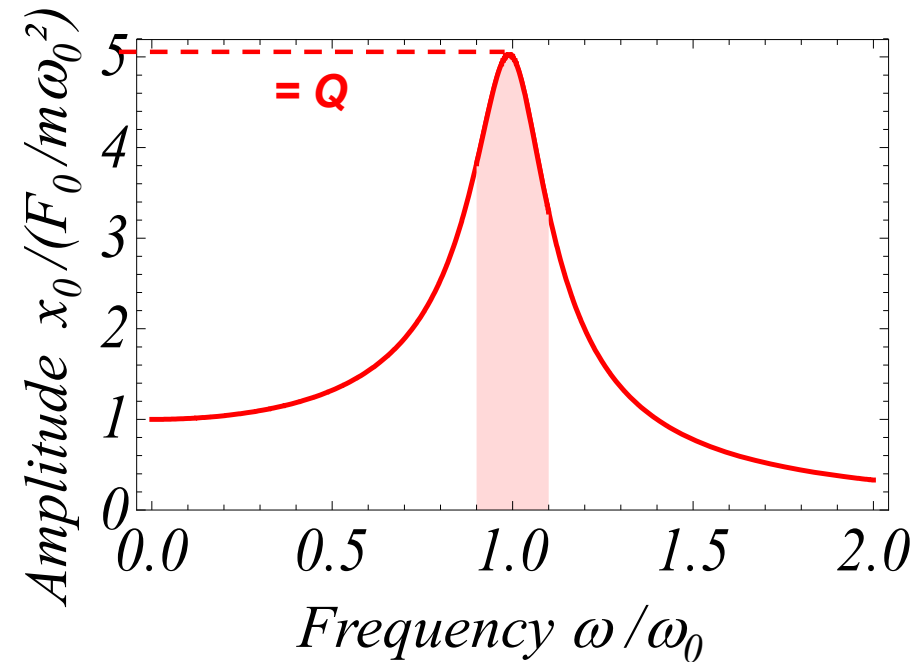
- quality factor: $Q = 1/(2\zeta)$
- bandwidth: $\Delta\omega = \omega_0/Q$

The damped forced harmonic oscillator (4) - Solution

$$x_0 = \frac{F_0}{m} \cdot \frac{1}{\sqrt{(\omega^2 - \omega_0^2)^2 + \frac{1}{Q^2} \omega^2 \omega_0^2}}$$

What did you learn?

- The displacement of a damped forced harmonic oscillator in the quasi stationary limit ($t \gg 1/\omega$) follows a sinusoidal curve with the amplitude x_0 .
- The displacement curve shows a magnification at resonance of Q compared to the low frequency limit.
- The resonance peak has a width of ω_0/Q .



Remark:

The magnification factor as well as the position of the 3dB-limits of the peak in the graph are slightly inaccurate. This is due to the here assumed poor **Q-factor of 5**. For $Q > 20$ you won't see a deviation from our results.

The damped forced harmonic oscillator (5) - Power

Input power

$$P_{in} = \dot{x}(t) \cdot F(t); \quad F(t) = -F_0 \cdot \sin \omega t; \quad x_{Res}(t) = -Q \cdot \frac{F_0}{m \cdot \omega_0^2} \cdot \cos \omega_0 t$$

$$\Rightarrow P_{in,Res} = Q \cdot \frac{F_0^2}{m \cdot \omega_0} \cdot \sin^2 \omega_0 t = Q \cdot \frac{F_0^2}{K} \cdot \omega_0 \cdot \sin^2 \omega_0 t$$

$$\Rightarrow \overline{P_{in,Res}} = Q \cdot \frac{F_0^2}{K} \cdot \omega_0 \cdot \left(\frac{\omega_0}{2\pi} \int_0^{\frac{2\pi}{\omega_0}} \sin^2 \omega_0 t \cdot dt \right) = \frac{1}{2} \cdot Q \cdot \frac{F_0^2}{K} \cdot \omega_0$$



The average input power resulting from the external force F is proportional to the quality factor Q .

The damped forced harmonic oscillator (6) - Power

Dissipated power

$$P_{diss} = \dot{x}(t) \cdot F_{damp}(t); \quad F_{damp}(t) = -c \cdot \dot{x}(t); \quad x_{Res}(t) = -Q \cdot \frac{F_0}{m \cdot \omega_0^2} \cdot \cos \omega_0 t$$

$$\Rightarrow P_{diss, Res} = -c \cdot Q^2 \cdot \frac{F_0^2}{m^2 \cdot \omega_0^2} \sin^2 \omega_0 t \quad \overset{c = \frac{m\omega_0}{Q}, \omega_0 = \sqrt{\frac{K}{m}}}{=} -Q \cdot \frac{F_0^2}{K} \omega_0 \cdot \sin^2 \omega_0 t = -P_{in, Res}$$

$$\Rightarrow \overline{P_{diss, Res}} = -\overline{P_{in, Res}}$$



The power dissipated in the damper exactly equals the input power (This quite obvious from energy conservation arguments).

The damped forced harmonic oscillator (7) - Power

Power to accelerate the mass

$$P_{inertia} = \dot{x}(t) \cdot F_{inertia}(t); \quad F_{inertia}(t) = -m \cdot \ddot{x}(t); \quad x_{Res}(t) = -Q \cdot \frac{F_0}{m \cdot \omega_0^2} \cdot \cos \omega_0 t$$

$$\Rightarrow P_{inertia, Res} = -Q^2 \cdot \frac{F_0^2}{m \cdot \omega_0} \sin \omega_0 t \cdot \cos \omega_0 t = -Q^2 \cdot \frac{F_0^2}{K} \cdot \omega_0 \sin \omega_0 t \cdot \cos \omega_0 t$$

$$\Rightarrow \overline{P_{inertia, Res}} = 0$$



The average kinetic energy of the system does not change. Nevertheless the instantaneous power required to accelerate the mass is huge (proportional Q^2).

The damped forced harmonic oscillator (8) - Power

Power to stretch the spring

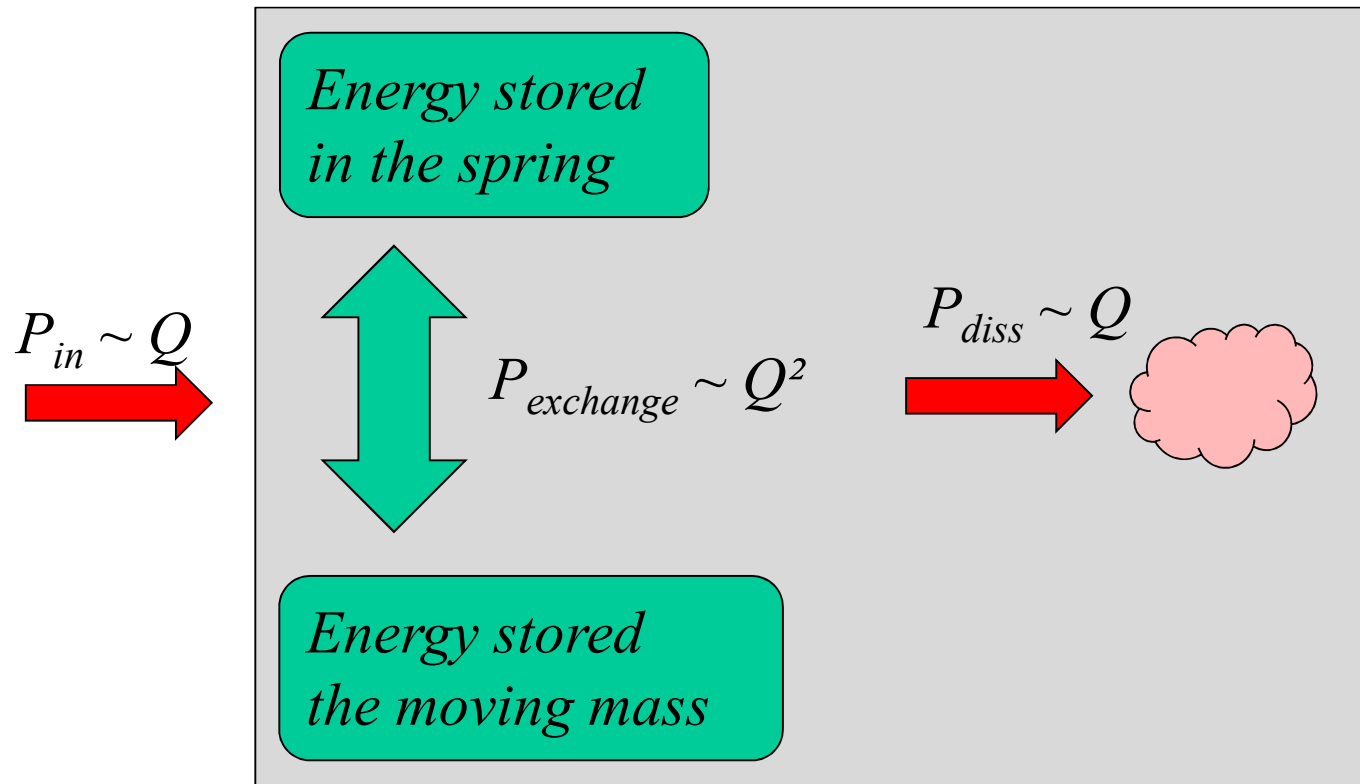
$$P_{pot} = \dot{x}(t) \cdot F_{pot}(t); \quad F_{pot}(t) = -K \cdot x(t); \quad x_{Res}(t) = -Q \cdot \frac{F_0}{m \cdot \omega_0^2} \cdot \cos \omega_0 t$$
$$\Rightarrow P_{pot, Res} = Q^2 \cdot \frac{K \cdot F_0^2}{m^2 \cdot \omega_0^3} \sin \omega_0 t \cdot \cos \omega_0 t = Q^2 \cdot \frac{F_0^2}{K} \cdot \omega_0 \sin \omega_0 t \cdot \cos \omega_0 t = -P_{inertia, Res}$$
$$\Rightarrow \overline{P_{pot, Res}} = 0$$



The power is transferred from the inertial to the potential system and back, but the sum of both is constant.

The damped forced harmonic oscillator (9) - Power

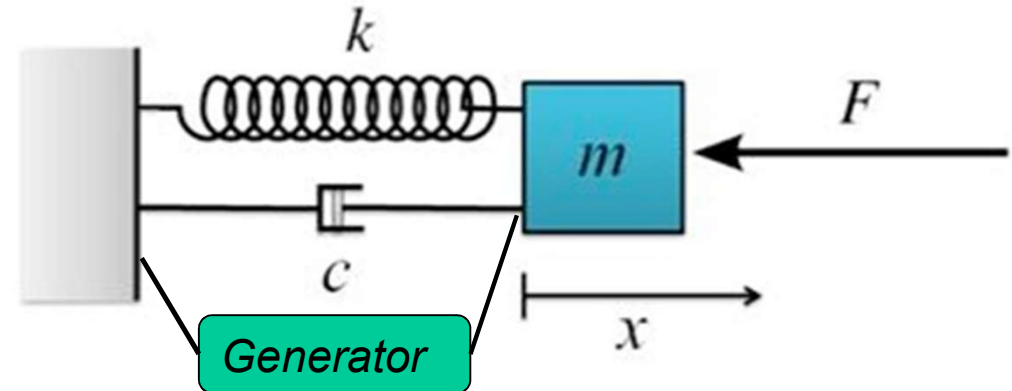
What did you learn ?



Vibrational Energy Harvesting (1)

What is vibrational Energy Harvesting ?

- The basic idea of vibrational energy harvesting is to attach a mechanism to a vibrating structure that converts the movement of the mass to electrical energy.
- To simplify our discussion we will always assume, that the generating mechanism behaves like the damper we have already discussed with the exception that it does not dissipate mechanical energy but converts it to useful electrical energy. This simplification often leads to quite accurate results.



Modeling vibrational energy harvesters

- Use exactly the same methods as in the previous slides but use $c_{\text{tot}} = c_{\text{harv}} + c_{\text{diss}}$ instead of c .
- Divide the total power to the damper into two parts, the generated power proportional to $c_{\text{harv}}/c_{\text{tot}}$ and the dissipated power proportional to $c_{\text{diss}}/c_{\text{tot}}$

Vibrational Energy Harvesting (2) - Power

Harvested power (in resonance)

$$P_{el} = \dot{x}(t) \cdot F_{harv}(t); \quad F_{harv}(t) = -c_{harv} \cdot \dot{x}(t); \quad x(t) = -Q \cdot \frac{F_0}{m \cdot \omega_0^2} \cdot \cos \omega_0 t$$

$$\Rightarrow P_{el} = -c_{harv} \cdot Q^2 \cdot \frac{F_0^2}{m^2 \cdot \omega_0^2} \sin^2 \omega_0 t$$

$$= -\frac{c_{harv}}{(c_{harv} + c_{dis})^2} \cdot F_0^2 \cdot \sin^2 \omega_0 t \quad \text{with } c_{harv} + c_{dis} = \frac{m\omega_0}{Q}, \omega_0 = \sqrt{\frac{K}{m}}$$

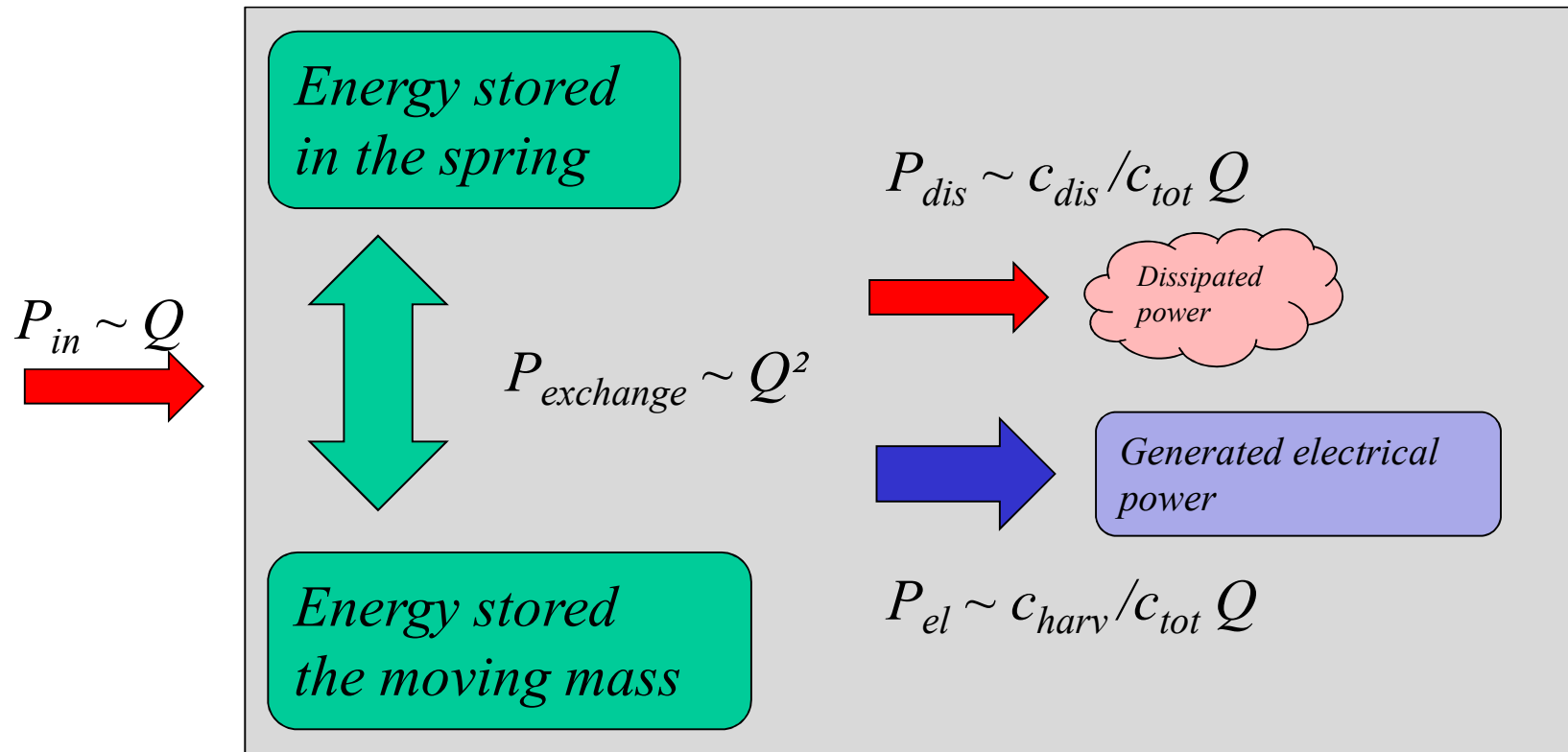
$$= -\frac{c_{harv}}{c_{harv} + c_{dis}} \cdot Q \cdot \frac{F_0^2}{m \cdot \omega_0} \cdot \sin^2 \omega_0 t$$

$$P_{el,avg} = -\frac{1}{2} \cdot \frac{c_{harv}}{(c_{harv} + c_{dis})^2} \cdot F_0^2$$



The harvested power depends on the ratio of c_{harv} and c_{dis}

Vibrational Energy Harvesting (3) - Power



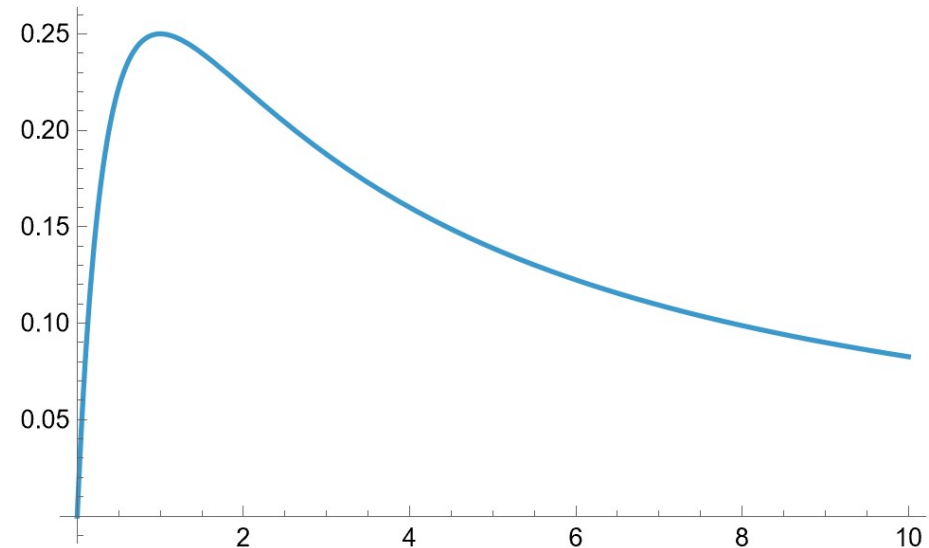
Vibrational Energy Harvesting (4) - Optimization

Given: a device with a dissipation c_{diss}

The output power can be expressed in terms of c_{diss} and c_{harv}

$$P_{el} \propto \frac{c_{harv}}{c_{tot}} Q_{tot} = \frac{c_{harv}}{c_{harv} + c_{diss}} \cdot \frac{2m\omega_0}{c_{harv} + c_{diss}}$$
$$= \frac{x}{(x+1)^2} \cdot \frac{2m\omega_0}{c_{diss}}$$

$$\text{with } x = \frac{c_{harv}}{c_{diss}}$$



maximum power is harvested for $x = 1$,
or $c_{diss} = c_{harv}$

Vibrational Energy Harvesting (5) - Optimization

Example

Case	C_{el}	$x/(1+x)^2$	P_{el}
very low generator coupling	$c_{harv} = 0.2 c_{diss}$	0.14	$0.56 P_{max}$
low generator coupling	$c_{harv} = 0.5 c_{diss}$	0.22	$0.89 P_{max}$
matched generator coupling	$c_{harv} = c_{diss}$	0.25	P_{max}
high generator coupling	$c_{harv} = 2 c_{diss}$	0.22	$0.89 P_{max}$
Very high generator coupling	$c_{harv} = 5c_{diss}$	0.14	$0.56 P_{max}$

What did you learn?

- The harvested power is optimized, if the damping constant for harvesting equals the damping constant for losses. $c_{harv} = c_{dis}$
- In that case half of the power transferred to the harvester is available as electrical energy
- The quality factor Q is reduced to half of its value compared to case with no harvesting
- A slight mismatch does not matter (factor of 2 mismatch leads to 11% loss).

Vibrational Energy Harvesting (6) - Power

So far:

$$P_{el}(\omega, c_{harv}) = -\frac{1}{2} \cdot c_{harv} \cdot \frac{1}{(\omega^2 - \omega_0^2)^2 + \left(\frac{c_{el} + c_{dis}}{m\omega_0}\right)^2 \omega^2 \omega_0^2} \cdot F_0^2$$

What is F_0 ?

The amplitude of inertial force: $F_0 = m \cdot a_{vib} = m \cdot \omega^2 \cdot x_{vib}$

$$P_{el}(\omega, c_{harv}) = -\frac{1}{2} \cdot c_{harv} \cdot \frac{m^2}{(\omega^2 - \omega_0^2)^2 + \left(\frac{c_{el} + c_{dis}}{m\omega_0}\right)^2 \omega^2 \omega_0^2} \omega^4 x_{vib}^2$$

$$P_{el}(\omega, c_{harv}) = -\frac{1}{2} \cdot c_{harv} \cdot \frac{m^2}{(\omega^2 - \omega_0^2)^2 + \left(\frac{c_{el} + c_{dis}}{m\omega_0}\right)^2 \omega^2 \omega_0^2} a_{vib}^2$$

Formulary for vibrational energy harvesting

Available average power:

$$P_{el}(\omega, c_{harv}) = \frac{1}{2} \cdot c_{harv} \cdot \frac{\omega^4}{(\omega^2 - \omega_0^2)^2 + \left(\frac{c_{el} + c_{dis}}{m\omega_0}\right)^2 \omega^2 \omega_0^2} a_{vib}^2$$

At resonance:

$$P_{el}(c_{harv}) = \frac{1}{2} \cdot \frac{c_{harv}}{(c_{harv} + c_{dis})^2} \cdot m^2 \cdot a_{vib}^2$$

With matched generator:

$$P_{el,max} = \frac{1}{8} \cdot \frac{1}{c_{dis}} \cdot m^2 \cdot a_{vib}^2$$

$$P_{el,max} = \frac{1}{8} \cdot \frac{1}{c_{dis}} \cdot m^2 \cdot \omega_0^4 \cdot x_{vib}^2$$

Vibrational Energy Harvesting (7) – How to..

How to choose a vibrational energy harvester?

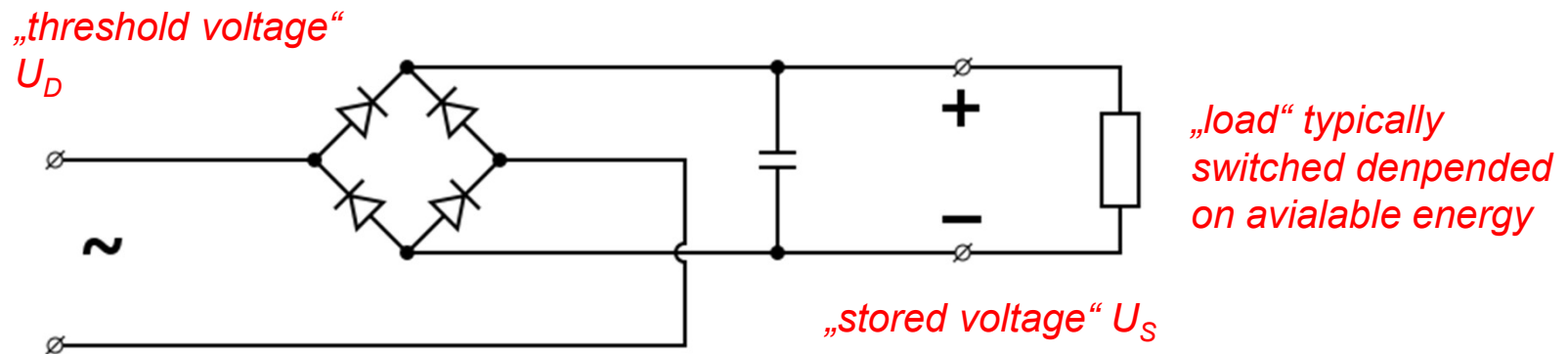
$$P_{el,max} = \frac{1}{8} \cdot \frac{1}{c_{dis}} \cdot m^2 \cdot a_{vib}^2$$

$$P_{el,max} = \frac{1}{8} \cdot \frac{1}{c_{dis}} \cdot m^2 \cdot \omega_0^4 \cdot x_{vib}^2$$

1. Find the vibration mode with the highest energy content, i.e with maximum $\omega^4 \cdot x_{vib}^2 \left(\frac{a_{vib}^2}{\omega^2}\right)$
2. Can we build a harvester with high mass m while still achieving resonance $\omega_0 = \sqrt{\frac{K}{m}}$?
3. Minimize any parasitic damping c_{dis}
4. Check for mechanical stability (resonance catastrophe)
5. Can we match the generator ? $c_{harv} = c_{dis}$
6. Check, if there is enough power for your application. $P_{el,max} = \frac{1}{8} \cdot \frac{1}{c_{dis}} \cdot m^2 \cdot \omega_0^4 \cdot x_{vib}^2$

The bridge rectifier (1)

In an energy-autonomous system you typically want to rectify the AC current from the harvester and smoothen the rectified AC current with a capacitor. The simplest way to do it is a bridge rectifier.



We will not go into detail:

- Assume another 50% loss for rectification.
- The voltage amplitude of the generator should be about $2U_S$ (U_S stored voltage under operating conditions).

Summary

What did you learn?

- You understand the concept of resonance for vibrational energy harvesting
- You can optimize the generator for maximum output
- You have a formulary for the harvested power under different conditions
- You are aware of the limitations of this approach especially the need of rectification and storage of the harvested power